

if-elif else Practice Questions

1. Write a program to check if a person is eligible to vote. Assume the minimum voting age is 18.
2. Write a program that takes marks of a student as input and checks if the student has passed or failed. Assume the passing mark is 40.
3. Write a program that takes two numbers as input and prints the larger number.
4. Write a program that takes marks as input and assigns a grade based on the following criteria:
 - Marks ≥ 90 : Grade A
 - Marks ≥ 80 : Grade B
 - Marks ≥ 70 : Grade C
 - Marks ≥ 60 : Grade D
 - Marks < 60 : Fail
5. Write a program that reads a number and categorizes it as "Negative", "Zero", or "Positive" using `if-elif-else`.
6. Write a program that takes two numbers and a choice (1 for addition, 2 for subtraction, 3 for multiplication, 4 for division) from the user and performs the corresponding operation.
7. Write a program that checks if a given year is a leap year or not using `if-elif-else`. A year is a leap year if it is divisible by 4 but not by 100, except when it is divisible by 400.

Loop Practice Questions

1. Write a program using a while loop to reverse a given number (e.g., input 1234, output 4321).
2. Write a program using a while loop to find the sum of all even numbers between 1 and 50.
3. Write a program using a for loop to calculate the factorial of a given number.
4. Write a program using a while loop to calculate the factorial of a given number.
5. Write a program using a for loop to print the first 10 terms of the Fibonacci series.
6. Write a program using a while loop to print the first 10 terms of the Fibonacci series.

Answers:

1.

Input number

num = 1234

reversed_num = 0

While loop to reverse the number

while num > 0:

digit = num % 10 # Get the last digit

reversed_num = reversed_num * 10 + digit # Append the digit

num //= 10 # Remove the last digit

print("Reversed Number:", reversed_num)

2.

Initialize variables

num = 1

sum_even = 0

While loop to calculate the sum of even numbers

while num <= 50:

if num % 2 == 0:

sum_even += num

num += 1

print("Sum of all even numbers between 1 and 50:", sum_even)

3.

Input number

num = 5

factorial = 1

For loop to calculate the factorial

for i in range(1, num + 1):

factorial *= i

print(f'Factorial of {num} is:', factorial)

4.

Input number

num = 5

factorial = 1

i = 1

While loop to calculate the factorial

while i <= num:

factorial *= i

i += 1

print(f'Factorial of {num} is:', factorial)

5.

Number of terms

terms = 10

a, b = 0, 1

For loop to print Fibonacci series

print("Fibonacci Series:", end=" ")

for _ in range(terms):

print(a, end=" ")

a, b = b, a + b

6.

Number of terms

terms = 10

a, b = 0, 1

count = 0

While loop to print Fibonacci series

print("Fibonacci Series:", end=" ")

while count < terms:

print(a, end=" ")

a, b = b, a + b

count += 1

Nested for loop: Pattern questions

1. Write a program using nested for loops to print the following pattern:

```
* * * * *
* * * * *
* * * * *
* * * * *
```

2. Write a program using nested for loops to print the following pyramid:

```
*
* *
* * *
* * * *
* * * * *
```

3. Write a program using nested for loops to print the following number pyramid:

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

4. Write a program using nested for loops to print the following alphabet pattern:

```
A
B B
C C C
D D D D
E E E E E
```

5. Write a program using nested for loops to print the following inverted number pyramid:

```
* * * * *
* * * *
* * *
* *
*
```

6. Write a program using nested for loops to print the following inverted number pyramid:

```
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

7. Write a program using nested for loops to print the following right-aligned triangle pattern:

```
  *
 * *
* * *
```

Solutions:

1.

Number of rows and columns

rows = 4

columns = 5

Outer loop for rows

for i in range(rows):

Inner loop for columns

for j in range(columns):

print("*", end=" ")

Move to the next line after printing each row

print()

2.

Number of rows

rows = 5

Outer loop for each row

for i in range(1, rows + 1):

Inner loop for printing asterisks

for j in range(i):

print("*", end=" ")

Move to the next line after each row

print()

3.

Number of rows

rows = 5

Outer loop for each row

for i in range(1, rows + 1):

Inner loop for printing numbers

for j in range(i):

print(i, end=" ")

Move to the next line after each row

print()

4.

```
# Number of rows
rows = 5

# Outer loop for each row
for i in range(rows):
    # Inner loop for printing letters
    for j in range(i + 1):
        # Print the corresponding alphabet, starting from 'A'
        print(chr(65 + i), end=" ")
    # Move to the next line after each row
    print()
```

5.

```
# Number of rows
rows = 5

# Outer loop for each row
for i in range(rows, 0, -1):
    # Inner loop for printing asterisks
    for j in range(i):
        print("*", end=" ")
    # Move to the next line after each row
    print()
```

6.

```
# Number of rows
rows = 5

# Outer loop for each row
for i in range(rows, 0, -1):
    # Inner loop for printing numbers
    for j in range(i):
        print(i, end=" ")
    # Move to the next line after each row
    print()
```

7.

```
# Number of rows
rows = 3

# Outer loop for each row
for i in range(1, rows + 1):
    # Inner loop for printing spaces
    for j in range(rows - i):
        print(" ", end=" ")
```

```
# Inner loop for printing asterisks
for k in range(i):
    print("*", end=" ")
# Move to the next line after each row
print()
```